

Vestibulo-ocular reflex (VOR)

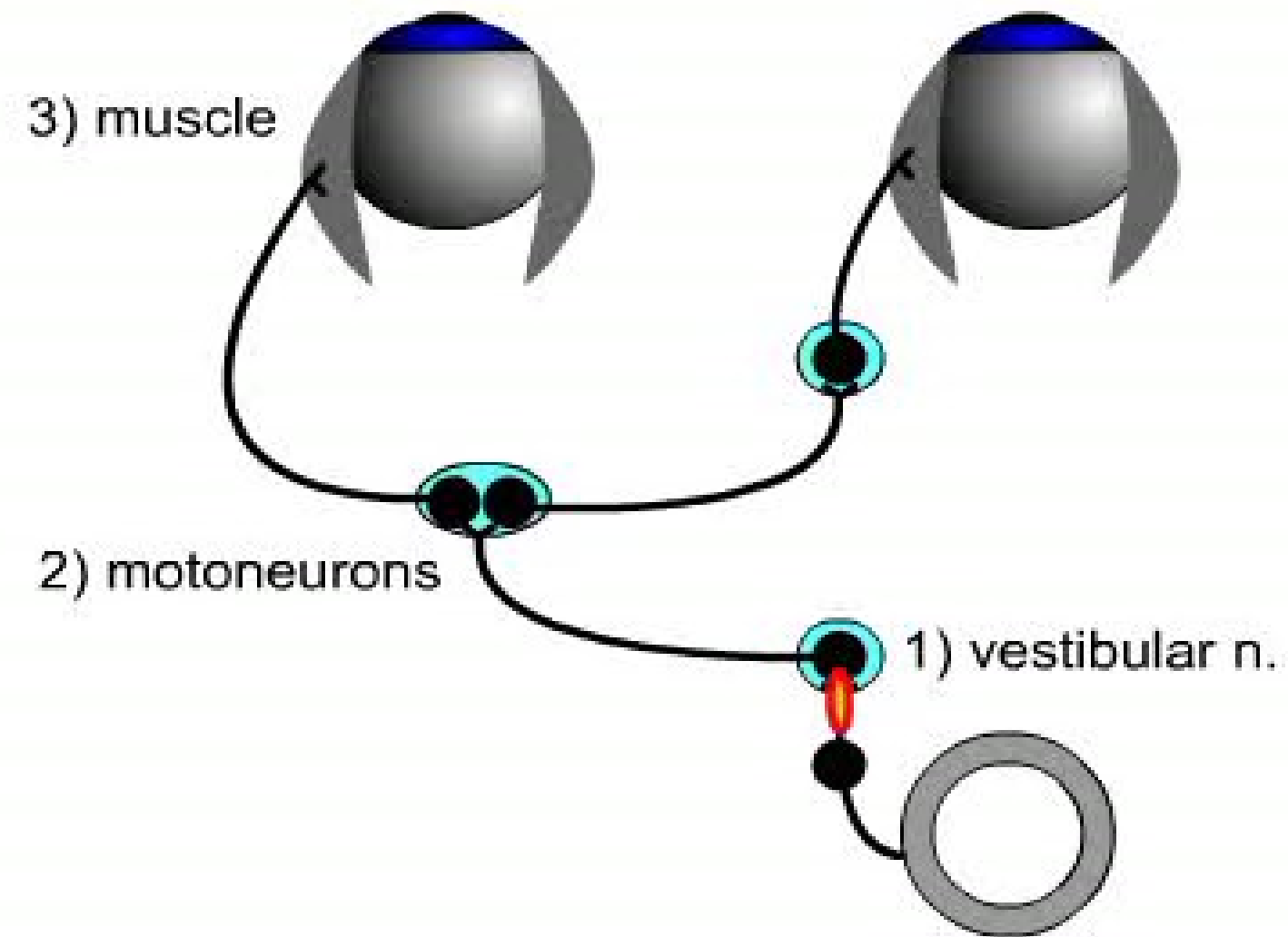
Consider the vestibulo-ocular reflex or head - video



Vestibulo-Ocular Reflex (VOR)

Explain the neural mechanism for a horizontal VOR.

The direct path is a short reflex with 3 synapses.



Start Labyrinth Otoliths Canals VOR Dizzy Plasticity End



Connect to iCub Simulator ports

/icubSim/cam/left

For receiving images from left camera

RPC port /icubSim/world

Create static sphere:

```
world mk ssph (radius) (three params for  
pos) (three params for colour)
```

```
world mk ssph 0.04 0.0 0.9 0.8 1 0 0
```

Change position of the static sphere

```
world set ssph (num) x y z
```

```
world set ssph 1 value1 value2 value3
```

http://wiki.icub.org/wiki/Simulator_README



Exercise Vestibulo-ocular reflex – Control the iCub simulator

Run the iCub Simulator (iCub_SIM)

PROGRAM 1:

- Create a static sphere;
- **For three iterations:**
 - Connect to the head of the iCub simulator and control joints 0 and 2 and make them moving with a sinusoidal input whose amplitude is 16 degrees and a frequency (that changes in each iteration of 0.1, 0.2 and 0,4 Hz). Each iteration lasts 15 seconds.
 - Ask the PROGRAM 2 to reset the joints 3 and 4 to zero value, wait for 5 seconds before starting the head motion;
- At the end of the 3 iterations, it sends a stop signal, wait 5 seconds and then terminate



Exercise VOR – Control the iCub simulator

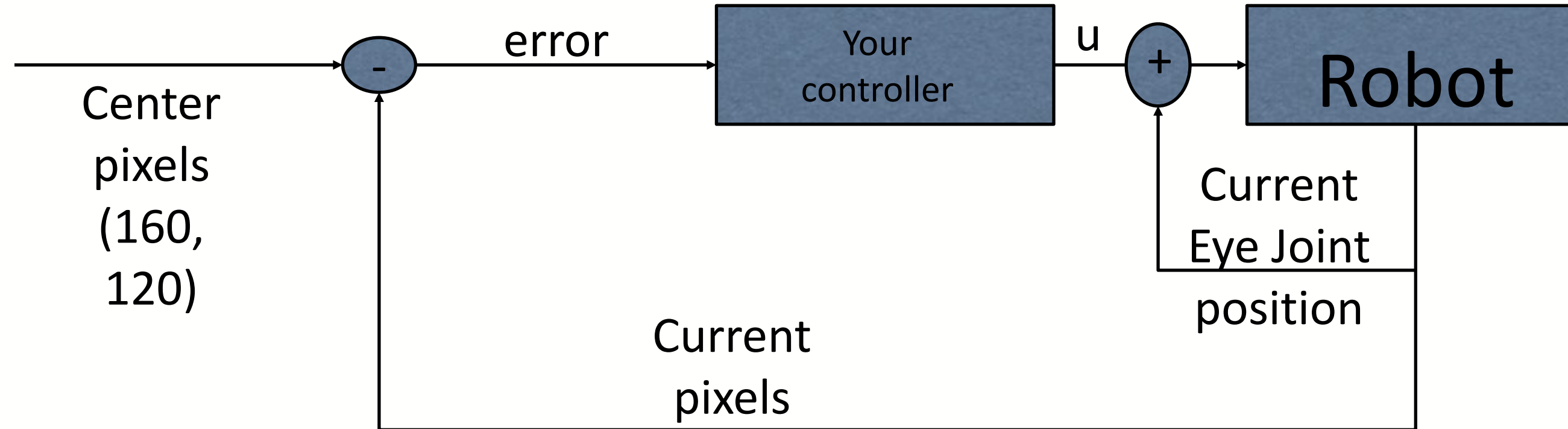
PROGRAM 2:

- Reset to 0 joint positions of the eye (joint 3 and 4) when required
- Receive images from the left camera
- Compute the pixels of the centre of the sphere
- Read Encoders values
- Send commands to joints 3 and 4 to move in position or velocity
- Terminate when required



Exercise VOR – Control the iCub simulator (PROGRAM 2)

You should make a closed loop controller in PROGRAM 2 as follows:



The goal is to keep the static sphere at the center of the image!



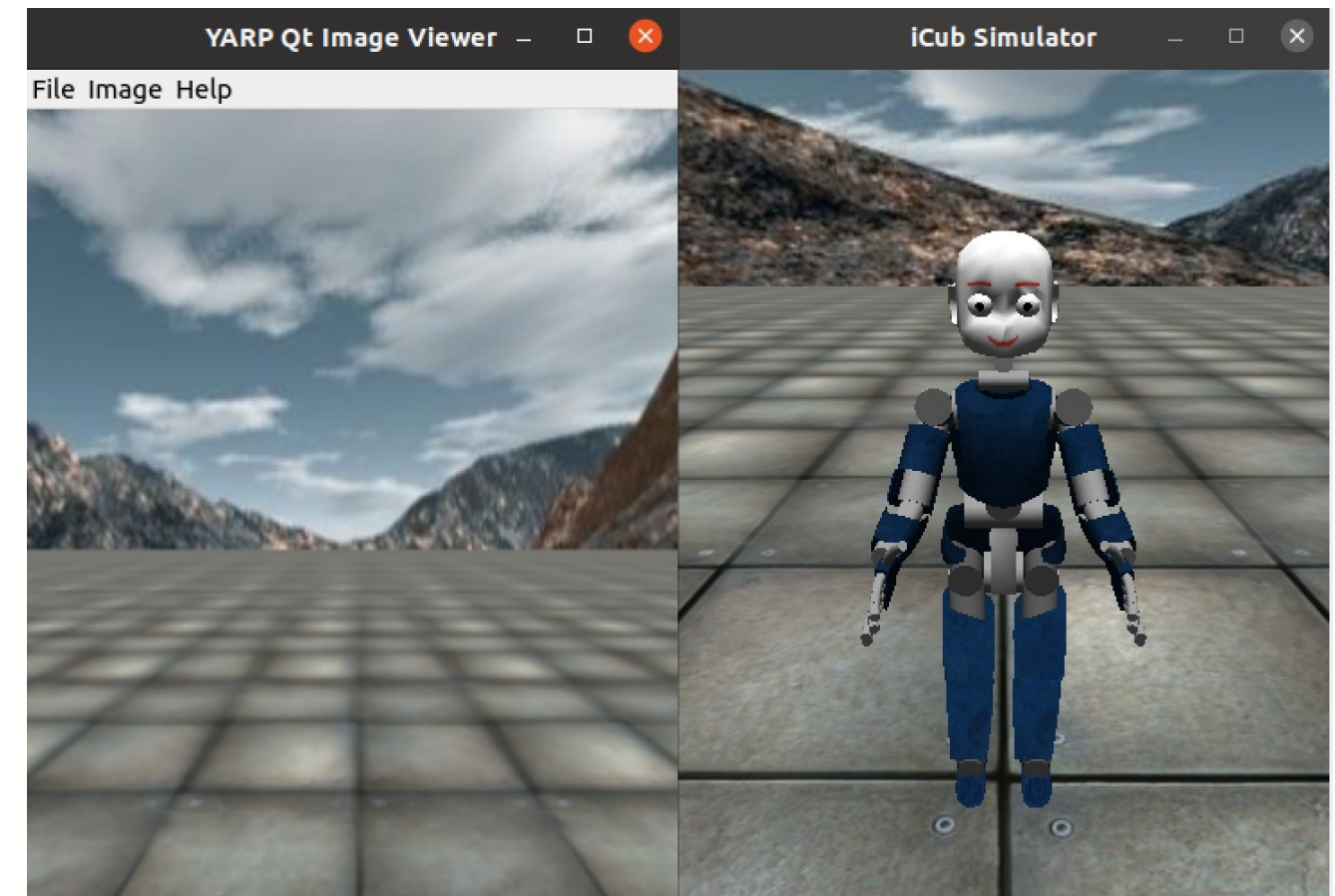
Exercise – VOR

Prepare a video when you run your program that shows, at the same time, the iCub simulator and the left camera view.

To enable the left camera view, you can use the following commands:

```
roscourse@roscourse-VirtualBox:~$ yarpview --name /viewer &
```

```
roscourse@roscourse-VirtualBox:~$ yarp connect /icubSim/cam/left /viewer
```



Make **graphs** for two iterations that show camera error (pixel), eye position (degrees) and head position (degrees) for the movement directions





MERRY
CHRISTMAS